

ABDELRAHMAN GAMAL ZAYED

AI & Machine Learning Engineer | Artificial Intelligence Engineering Student

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[LinkedIn](#) · [GitHub](#) · [Portfolio](#)

PROFESSIONAL SUMMARY

Artificial Intelligence Engineering student seeking an AI Engineer Internship with hands-on experience in machine learning, deep learning, computer vision, and natural language processing. Proficient in Python, TensorFlow, PyTorch, Scikit-learn, YOLO, LangChain, and Hugging Face Transformers. Delivered end-to-end AI solutions across three concurrent internships covering computer vision, applied AI (NLP, LLMs, RAG), and agentic AI systems. Co-author of a peer-reviewed research paper on hybrid metaheuristic forecasting models.

PROFESSIONAL EXPERIENCE

Agentic AI Trainee | Digital Egypt Pioneers Initiative (DEPI) Jul 2026 – Present

- Built multi-agent workflows using LangChain & LangGraph with autonomous planning, tool-calling, and memory modules; deployed RAG pipelines (FAISS, Chroma) for knowledge-grounded LLM responses.
- Evaluated agent safety and reliability across structured benchmarks, applying prompt engineering and chain-of-thought techniques to maximise task completion accuracy.

Applied AI Trainee (NLP · CV · LLMs · RAG) | Information Technology Institute (ITI) Jul 2026 – Aug 2026

- Fine-tuned BERT/GPT transformer models for NLP tasks (classification, NER, sentiment); engineered RAG pipelines linking LLM backends to curated knowledge bases for enterprise Q&A.
- Applied YOLOv8/v11 for real-time object detection on domain-specific datasets, optimising inference pipelines from training to deployment-ready export.

Computer Vision Intern | National Telecommunication Institute (NTI) Jul 2026 – Aug 2026

- Delivered applied CV projects (traffic monitoring, defect detection) using pretrained ResNet/EfficientNet with custom fine-tuning; achieved top mAP via YOLO + AdamW optimiser tuning.
- Benchmarked models on precision, recall, and mAP@50; applied pruning & quantisation to meet edge-deployment constraints.

Deep Learning Trainee | Egyptian Engineers Syndicate Sep 2025

- Completed intensive training on deep neural networks, transfer learning, and regularisation; logged loss curves and experimentation architectures for systematic model improvement.

Machine Learning Intern | National Telecommunication Institute (NTI) Jul 2025

- Engineered a diabetes-risk classification pipeline (Logistic Regression, Random Forest, SVM, Decision Trees); minimised false negatives via stratified cross-validation and class-imbalance correction.

EDUCATION

B.Eng. in Artificial Intelligence Engineering Oct 2023 – Jul 2028

New Ismailia National University · Ismailia, Egypt · GPA: **3.7 / 4.0**

- Relevant Coursework: Machine Learning, Deep Learning, Data Science, Data Structures, OOP, Computer Science.

PROJECTS

Solar Power Forecasting – Hybrid Metaheuristic Model · Research · Co-Author Under Review

- Proposed ELM + Improved Harris Hawks Optimisation (IHO) hybrid framework; outperformed CatBoost, XGBoost & Transformer baselines on R^2 , RMSE, and MAPE metrics.

Facial Emotion Recognition · Deep Learning · Computer Vision

- 7-class CNN classifier (TensorFlow/Keras + OpenCV) with data augmentation and real-time validation monitoring.

Multilingual Fake News Classifier · NLP

- Bilingual Arabic/English text classification using custom preprocessing, tokenisation, and statistical feature extraction.

Diabetes Prediction Model · Machine Learning · Healthcare

- Risk classification pipeline with hyperparameter tuning, stratified cross-validation, and class-imbalance optimisation.

SKILLS

Languages & Tools

Python, C, C++, Jupyter, Git/GitHub, PyCharm

ML / DL

Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, CNNs, Transfer Learning

Computer Vision

YOLO (v5/v8/v11), ResNet, EfficientNet, VGG, Pretrained Models, Model Quantisation

NLP & LLMs

Hugging Face, BERT, GPT, LangChain, LangGraph, AutoGen, RAG, FAISS, Chroma

Core Algorithms

Logistic Regression, Random Forest, SVM, XGBoost, CatBoost, ELM, Transformers

Soft Skills

Leadership, Team Collaboration, Time Management, Communication, Analytical Problem-Solving, Adaptability

Languages

Arabic (Native) · English (Intermediate)

LEADERSHIP & VOLUNTEERING

AI Head | Microsoft Students Club – Suez Canal University (SCU) Present

- Direct AI-focused student workshops, hackathons, and peer learning pathways across ML, computer vision, and NLP domains.

Volunteer | Egyptian Youth Entity (EYE) Present

- Collaborate with organising committees to drive youth-led community development and social impact initiatives.